Exercise 1

I
$$a^T \cdot x = \sum_{i=0}^{n-1} a_i x_i$$
$$\frac{\partial a^T x}{\partial x_j} = \frac{\partial \sum_{i=0}^{n-1} a_i x_i}{\partial x_j} = a_j$$
$$\rightarrow \frac{\partial a^T x}{\partial x} = a^T$$

II.
$$a^T A a = a^T \cdot \left(\sum_{i=0}^{n-1} \left(\sum_{j=0}^{n-1} A_{ij} \cdot a_j \right)\right) = \sum_{i=0}^{n-1} a_i \cdot \left(\sum_{j=0}^{n-1} \cdot A_{ij} a_j \right) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} a_i \cdot A_{ij} a_j$$
$$\frac{\partial a^T A a}{\partial a_k} = \frac{\partial \left(\sum_{i=0}^{n-1} \sum_{j=0}^{n-1} a_i A_{ij} a_j \right)}{\partial a_k} = \frac{\partial \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} A_{ij} \cdot (a_i a_j)}{\partial a_k} = \sum_{j=0}^{n-1} A_{kj} a_j + \sum_{i=0}^{n-1} A_{ik} a_i$$
$$\rightarrow \frac{\partial a^T A a}{\partial a} = \sum_{k=0}^{n-1} \frac{\partial a^T A a}{\partial a_k} = \sum_{k=0}^{n-1} \left(\sum_{j=0}^{n-1} A_{kj} a_j + \sum_{i=0}^{n-1} A_{ik} a_i \right) = \sum_{k=0}^{n-1} \sum_{j=0}^{n-1} A_{kj} a_j + \sum_{k=0}^{n-1} \sum_{i=0}^{n-1} A_{ik} a_i = a^T A^T + a^T A = a^T (A^T + A) = a^T (A + A^T)$$

III.
$$\frac{\partial (x-As)^T (x-As)}{\partial s_j} = \frac{\partial \left(\sum_{k=0}^{n-1} (x-As)_k (x-As)_k \right)}{\partial s_j} = \frac{\partial \sum_{k=0}^{n-1} x_k^2 - 2 x_k (As)_k + (As)_k^2}{\partial s_j} = \frac{\partial \sum_{k=0}^{n-1} (As)_k^2 - 2 x_k (As)_k}{\partial s_j}$$
$$= \frac{\partial \sum_{k=0}^{n-1} \left(\sum_{i=0}^{n-1} A_{ki} s_i \right)^2 - 2 x_k \left(\sum_{i=0}^{n-1} A_{ki} s_i \right)}{\partial s_j} = 2 \cdot \sum_{k=0}^{n-1} \sum_{i=0}^{n-1} A_{ki} s_i A_{kj} - 2 \cdot \sum_{k=0}^{n-1} x_k \cdot A_{kj} = \sum_{k=0}^{n-1} -2 \cdot \left(x_k A_{kj} - \sum_{i=0}^{n-1} A_{ki} s_i A_{kj} \right)$$
$$= -2 \sum_{k=0}^{n-1} \left(x_k - \sum_{i=0}^{n-1} A_{ki} s_i \right) A_{kj} = \left(-2 \cdot (x-As)^T A \right)_j$$
$$\frac{\partial (x-As)^T (x-As)}{\partial s} = \sum_{i=0}^{n-1} \frac{\partial (x-As)^T (x-As)}{\partial s_i} = \sum_{i=0}^{n-1} \left(-2 (x-As)^T A \right)_i = -2 (x-As)^T A$$